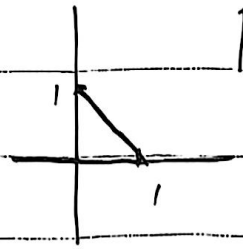
作业10.

$r=\frac{\sqrt{2}}{2}\sec(\frac{\pi}{4}-\theta),\ 0\le\theta\le\frac{\pi}{2}$

13.(3).




原式 $=\int_0^{\frac{\pi}{2}}d\theta\int_0^{\frac{\sqrt{2}}{2}\sec(\frac{\pi}{4}-\theta)}f(r\cos\theta,r\sin\theta)\,r\,dr.$

14.(3). $f=x^2+y^2=r^2.$

积分区域 $r^4\le a^2r^2(\cos^2\theta-\sin^2\theta)$

即 $r^2\le a^2\cos2\theta.$

$\therefore 0\le r\le\sqrt{a^2\cos2\theta},\ \theta\in[0,\frac{\pi}{4}]\cup[\frac{3}{4}\pi,\frac{5}{4}\pi]\cup[\frac{7}{4}\pi,2\pi]$

$\therefore$ 原式 $=2\int_{\frac{3}{4}\pi}^{\frac{5}{4}\pi}d\theta\int_0^{\sqrt{a^2\cos2\theta}}r^3dr$

$=\frac{a^4}{2}\int_{\frac{3}{4}\pi}^{\frac{5}{4}\pi}(\cos^2\theta)\,d\theta=\frac{a^4}{2}\cdot\frac{\pi}{4}=\frac{1}{8}\pi a^4$

(4). $f=\sqrt{\frac{1-r^2}{1+r^2}}$，积分区域为 $r=1,\ 0\le\theta\le\frac{\pi}{2}$

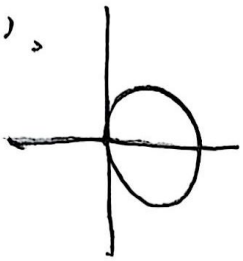
$\therefore$ 原式 $=\int_0^{\frac{\pi}{2}}d\theta\int_0^1 r\sqrt{\frac{1-r^2}{1+r^2}}\,dr$ 令 $r^2=t$.

$\therefore$ 原式 $=\frac{1}{2}\int_0^{\frac{\pi}{2}}d\theta\int_0^1\sqrt{\frac{1-t}{1+t}}\,dt$ 令 $t=\sin x$.

$\therefore$ 原式 $=\frac{1}{2}\int_0^{\frac{\pi}{2}}d\theta\int_0^{\frac{\pi}{2}}(1-\sin x)\,dx$

$=\frac{1}{2}\cdot\frac{\pi}{2}\cdot(\frac{\pi}{2}-1)=\frac{1}{8}\pi^2-\frac{1}{4}\pi$

15.(1).



$\theta = \arccos(\frac{r}{\sqrt{2}a})$.

原式 $= \int_0^1 dr \int_{-\arccos(\frac{r}{\sqrt{2}a})}^{\arccos(\frac{r}{\sqrt{2}a})} f(r,\theta)\, d\theta$

16.(1). 原式 $= \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r e^{r^2} dr$

$= \frac{1}{2}\int_0^{\frac{\pi}{2}} (e-1)\, d\theta = \frac{\pi}{4}(e-1)$

(3). 原式 $= \int_0^{\pi} d\theta \int_0^2 r^5 \cos^2\theta \sin^2\theta\, dr$

$= \frac{8}{3}\int_0^{\pi} \sin^2 2\theta\, d\theta = \frac{8}{3}\int_0^{\pi} (\frac{1-\cos 2\theta}{2}) = \frac{4}{3}\pi$.

(6). 左半: $x = 1+\sqrt{1-y^2} \Rightarrow (x-1)^2+y^2=1$

右半: $x = \sqrt{2y-y^2} \Rightarrow x^2+(y-1)^2=1$

$\therefore$ 令 $f = \exp(\frac{xy}{x^2+y^2})$

$= \exp(\frac{1}{2}\sin 2\theta)$.

$\therefore$ 原式 $= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_{2\cos\theta}^{2\sin\theta} r\exp(\frac{1}{2}\sin 2\theta)\, dr$.

$= \int_0^{\frac{\pi}{4}} d\theta \int_{2\sin\theta}^{2\cos\theta} r\exp(\frac{1}{2}\sin 2\theta)\, dr$,

$= \frac{1}{2}\int_0^{\frac{\pi}{4}} \cos 2\theta \exp(\frac{1}{2}\sin 2\theta)\, d\theta = 2\sqrt{e}-2$.

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17.

(1). 令 $u=3x,\ v=2y$ $\therefore x=\frac{1}{3}u,\ y=\frac{1}{2}v.$

$\therefore |J| = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \frac{1}{6}$

$\therefore$ 原式 $=\frac{1}{6}\iint_D \sin(u^2+v^2)\,du\,dv$ 积分区域为 $u^2+v^2\le 1,\ u\ge 0,\ v\ge 0.$

$= \frac{1}{6}\int_0^{\frac{\pi}{2}} d\theta \int_0^1 r\sin r^2\,dr = \frac{\pi}{24}(1-\cos 1)$

(4). $|y|\le 1-|x|$ $\therefore -1+|x|\le y\le 1-|x|.$

$\therefore$ 原式 $=\int_{-1}^{1}dx\int_{-1+|x|}^{1-|x|} e^{x+y}\,dy.$

$=\int_{-1}^{1} e^x\left(e^{1-|x|}-e^{-1+|x|}\right)dx$

$=\int_0^1 (e-e^{-1+2x})\,dx+\int_{-1}^{0}(e^{1+2x}-e^{-1})\,dx$

$= e-e^{-1}$

(5). 令 $u=x+y,\ v=x-y$ $\therefore x=\frac{1}{2}(u+v),\ y=\frac{1}{2}(u-v).$

$\therefore (u,v)$ 平面上顶点为

$(\pi,\pi),\ (3\pi,\pi),\ (5\pi,-\pi),\ (\pi,-\pi).$

$J=\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & -\frac{1}{2}\end{vmatrix} = -\frac{1}{2},\ |J|=\frac{1}{2}.$

$\therefore$ 原式 $=\frac{1}{2}\int_{\pi}^{5\pi}du\int_{-\pi}^{\pi}u^3\cos^2 v\,dv = 78\pi^5$

18.

(1). $x^2-y^2=uv$, $(x+y)^2=u^2$, $x=\frac{1}{2}(u+v)$, $y=\frac{1}{2}(u-v)$

$$|J|=\left|\frac{\partial(x,y)}{\partial(u,v)}\right|=\frac{1}{2}$$

积分区域 $D$: $\frac{1}{2}(u-v)\le\frac{1}{2}(u+v)\le\sqrt{1-\frac{1}{4}(u-v)^2}$

即: 令 $h(u)=\min\{u,\sqrt{2-u^2}\}$ $\therefore D: 0\le v\le h(u),\ 0\le u\le\sqrt{2}$

$\therefore$ 原式 $=\frac{1}{2}\int_0^{\sqrt{2}}du\int_0^{h(u)}uve^{u^2}dudv=\frac{1}{8}(e-1)^2$.

(2). $J=\dfrac{1}{\frac{\partial(u,v)}{\partial(x,y)}}=\dfrac{1}{2(x^2-y^2)}$

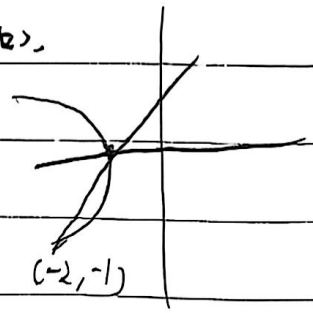
积分区域由 $\begin{cases}u=1 & \text{边界}\\ v=0.\\ v=\frac{1}{2}u\end{cases}$

即: $0\le u\le 1$; $0\le v\le\frac{1}{2}u$.

$(x+y)^2=u+2v$.

原式 $=\frac{1}{2}\int_0^1du\int_0^{\frac{1}{2}u}\exp(u+2v)\,dv=\frac{1}{8}(e-1)^2$.

19. (2).




$$A=\int_{-1}^{0}dy\int_{y-1}^{-1-y^2}dx$$

$$=-\int_{-1}^{0}(y^2+y)=\frac{1}{6}$$

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(4). $\because 4\sin\theta \geq 2$. $\therefore \frac{\pi}{6} \leq \theta \leq \frac{5}{6}\pi$

$$\therefore A = \int_{\frac{\pi}{6}}^{\frac{5}{6}\pi} d\theta \int_{2}^{4\sin\theta} r\,dr$$

$$= \int_{\frac{\pi}{6}}^{\frac{5}{6}\pi} (8\sin^2\theta - 2)\,d\theta = \frac{4}{3}\pi + 2\sqrt{3},$$

(6). 令 $x = r\cos\theta$, $y = r\sin\theta$,

$\therefore$ 原式: $r^4 = 2ar^3\cos^3\theta$

即: $0 \leq r \leq 2a\cos^3\theta$

$\because 2a\cos^3\theta \geq 0$ $\therefore -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$$\therefore A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_{0}^{2a\cos^3\theta} r\,dr$$

$$= 4a^2 \int_0^{\frac{\pi}{2}} \cos^6\theta = \frac{5}{8}\pi a^2.$$

(7). 令 $u = 2x + 3y + 4$, $v = 5x + 6y + 7$.

$$\therefore |J| = \frac{1}{\left|\frac{\partial(u,v)}{\partial(x,y)}\right|} = \frac{1}{3}$$

$$\therefore A = \frac{1}{3}\int_{-3}^{3} dx \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dy = \frac{1}{3}\cdot 9\pi = 3\pi.$$

20

11、设区域在 $y-z$ 平面上投影为 $D_{yz}$.

$\therefore V=\iint\limits_{D_{yz}} dy\,dz\int_0^{2y}dx$

$=2\iint\limits_{D_{yz}} y\,dy\,dz.$

取 $y=r\cos\theta$, $z=r\sin\theta$.

$\therefore$ 区域: $0\le\theta\le\frac{\pi}{2}$, $0\le r\le 2$.

$\therefore V=2\int_0^{\frac{\pi}{2}}\cos\theta\,d\theta\int_0^2 r^2dr=\frac{16}{3}.$

13、$xOy$ 平面上的投影: $x^2+y^2=1$. 记为 $D_{xy}$.

$\therefore V=\iint\limits_{D_{xy}} dx\,dy\int_{x^2+y^2}^{1+\sqrt{1-x^2-y^2}}dz$

$=\iint\limits_{D_{xy}}(\sqrt{1-x^2-y^2}+1-x^2-y^2)\,dx\,dy.$

令 $x=r\cos\theta$, $y=r\sin\theta$.

区域: $0\le\theta\le 2\pi$, $0\le r\le 1$

$\therefore V=\int_0^{2\pi}d\theta\int_0^1(\sqrt{1-r^2}+1-r^2)r\,dr$

$=2\pi\int_0^1(r-r^3+r\sqrt{1-r^2})dr=\frac{7}{6}\pi.$

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15、xOy投影区域为：$\begin{cases} x \geq 0, y \geq 0 \\ y \leq -x+1 \end{cases}$ 记为 $D_{xy}$，$0 \leq z \leq 6-(x^2+y^2)$

$$\therefore V = \iint_{D_{xy}} dxdy \int_0^{6-(x^2+y^2)} dz.$$

$$= \iint_{D_{xy}} (6-x^2-y^2)\,dxdy$$

$$= \int_0^1 dx \int_0^{-x+1} (6-x^2-y^2)\,dy = \frac{17}{6}.$$

21、$m = \int_0^1 dx \int_{2-x}^{x} (x^2+y^2)\,dy.$

$$= \int_0^1 \left(x^2(x-(2-x)) + \frac{1}{3}(x^3-(2-x)^3)\right) dx.$$

$$= \frac{4}{3}.$$

思考题

一、$V_n(R)=\int_{-R}^{R}\left[\int_{x_1^2+x_2^2+\cdots+x_{n-1}^2\le R^2-x_n^2} dx_1\cdots dx_{n-1}\right]dx_n$

$=\int_{-R}^{R}V_{n-1}(\sqrt{R^2-x_n^2})\,dx_n$

设 $V_{n-1}(R)=C_{n-1}R^{n-1}$

$\therefore V_n(R)=C_{n-1}\int_{-R}^{R}(R^2-x_n^2)^{\frac{n-1}{2}}dx_n$.

令 $t=\frac{x_n}{R}$ $\therefore V_n(R)=C_{n-1}R^n\int_{-1}^{1}(1-t^2)^{\frac{n-1}{2}}dt$.

即: $\frac{V_n(R)}{R^n}=C_{n-1}\int_{-1}^{1}(1-t^2)^{\frac{n-1}{2}}dt$, 与R无关

又 $\because V_n(1)=C_n$, 与R无关

$\therefore$ 由归纳法: $V_n(R)\propto R^n$.

二、$\frac{V_n(R)}{RV_{n-1}(R)}=\int_{-1}^{1}(1-t^2)^{\frac{n-1}{2}}dt=2\int_0^1(1-t^2)^{\frac{n-1}{2}}dt$.

令 $u=t^2$.

$\therefore$ 上式 $=\int_0^1 u^{-\frac{1}{2}}(1-u)^{\frac{n-1}{2}}du$

$=\frac{\Gamma(\frac{1}{2})\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n+2}{2})}$ $\because \Gamma(\frac{1}{2})=\sqrt{\pi}$.

$\therefore$ 上式 $=\frac{\sqrt{\pi}\,\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n+2}{2})}$

即: $V_n(R)=R\sqrt{\pi}\,\frac{\Gamma(\frac{n}{2}+\frac{1}{2})}{\Gamma(\frac{n}{2}+1)}V_{n-1}(R)$.

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三. 令 $r=\sqrt{x_{n-1}^2+x_n^2}$.

$\therefore V_n(R)=\int_0^R V_{n-2}(\sqrt{R^2-r^2})\,2\pi r\,dr$.

令 $t=\frac{r}{R}$

$\because V_{n-2}(R)=C_{n-2}R^{n-2}$.

$\therefore V_n(R)=2\pi C_{n-2}R^n\int_0^1 t(1-t^2)^{\frac{n-2}{2}}dt$.

令 $u=t^2$.

$\therefore \int_0^1 t(1-t^2)^{\frac{n-2}{2}}dt=\frac{1}{2}\int_0^1(1-u)^{\frac{n-2}{2}}du=\frac{1}{n}$

即: $V_n(R)=\frac{2\pi}{n}R^nV_{n-2}(R)$.

四. $V_0(R)=1$, $V_1(R)=2R$.

①. $V_n(R)=R\sqrt{\pi}\,\frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2}+1)}V_{n-1}(R)$

$=(R\sqrt{\pi})^2\,\frac{\Gamma(\frac{n+1}{2})\Gamma(\frac{n}{2})}{\Gamma(\frac{n}{2}+1)\Gamma(\frac{n-1}{2}+1)}V_{n-2}(R)$

$=\cdots$

$=R^n\pi^{\frac{n}{2}}\,\frac{\Gamma(\frac{n+1}{2})\cdots\Gamma(\frac{3}{2})}{\Gamma(\frac{n}{2}+1)\Gamma(\frac{n-1}{2}+1)\cdots\Gamma(1)}V_0(R)$.

$=\frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2}+1)}R^n$.

② $V_n(R)=\frac{2\pi}{n}R^2V_{n-2}(R)$

$n=2k$ 时：

$$V_{2k}(R)=\frac{2\pi}{2k}R^2\frac{2\pi}{(2k-2)}R^2\cdots\frac{2\pi}{2}R^2V_0(R)$$

$$=\frac{(2\pi)^k}{2^k k!}R^{2k}=\frac{\pi^k}{k!}R^{2k}=\frac{\pi^k}{\Gamma(k+1)}R^{2k}.$$

$n=2k+1$ 时.

$$V_{2k+1}(R)=\frac{2\pi}{2k+1}R^2\frac{2\pi}{2k-1}R^2\cdots\frac{2\pi}{3}R^2V_1(R).$$

$$=\frac{\pi^{k+\frac{1}{2}}}{\Gamma(k+\frac{3}{2})}R^{2k+1}$$

综合可：$V_n(R)=\frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2}+1)}R^n$

3. n维球坐标：

$$\begin{cases}x_1=r\cos\theta_1\\x_2=r\sin\theta_1\cos\theta_2\\\vdots\\x_n=r\sin\theta_1\cdots\sin\theta_{n-2}\sin\theta_{n-1}\end{cases}$$

$\therefore dV=r^{n-1}(\sin^{n-2}\theta_1)(\sin^{n-3}\theta_2)\cdots(\sin\theta_{n-2})d\theta_1\cdots d\theta_{n-1}$

$\therefore V_n(R)=\int_0^R r^{n-1}dr\cdot\int_0^{2\pi}d\theta_{n-1}\prod_{k=1}^{n-2}\int_0^{\pi}\sin^{(n-1-k)}(\theta_k)d\theta_k$

记角向积分为 $S$.

径向 $\int_0^R r^{n-1}dr=\frac{R^n}{n}$

$$S=(2\pi)\cdot(\sqrt{\pi})^{n-2}\frac{\Gamma(1)\Gamma(\frac{3}{2})\cdots\Gamma(\frac{n-1}{2})}{\Gamma(\frac{3}{2})\Gamma(2)\cdots\Gamma(\frac{n}{2})}$$

$$=\frac{2\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2})}$$

$$\therefore V_n(R)=\frac{R^n}{n}\cdot\frac{2\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2})}=\frac{R^n\cdot\pi^{\frac{n}{2}}}{\frac{n}{2}\Gamma(\frac{n}{2})}=\frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2}+1)}R^n$$

即: $V_n(R)=\frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2}+1)}R^n$ 也符合.